Lecture 8 Image Super Resolution

（Generative Models for Super Resolution）

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Outline

- Interpolation methods
- Reconstruction based methods
- Deep learning based methods
- Single-image Super Resolution

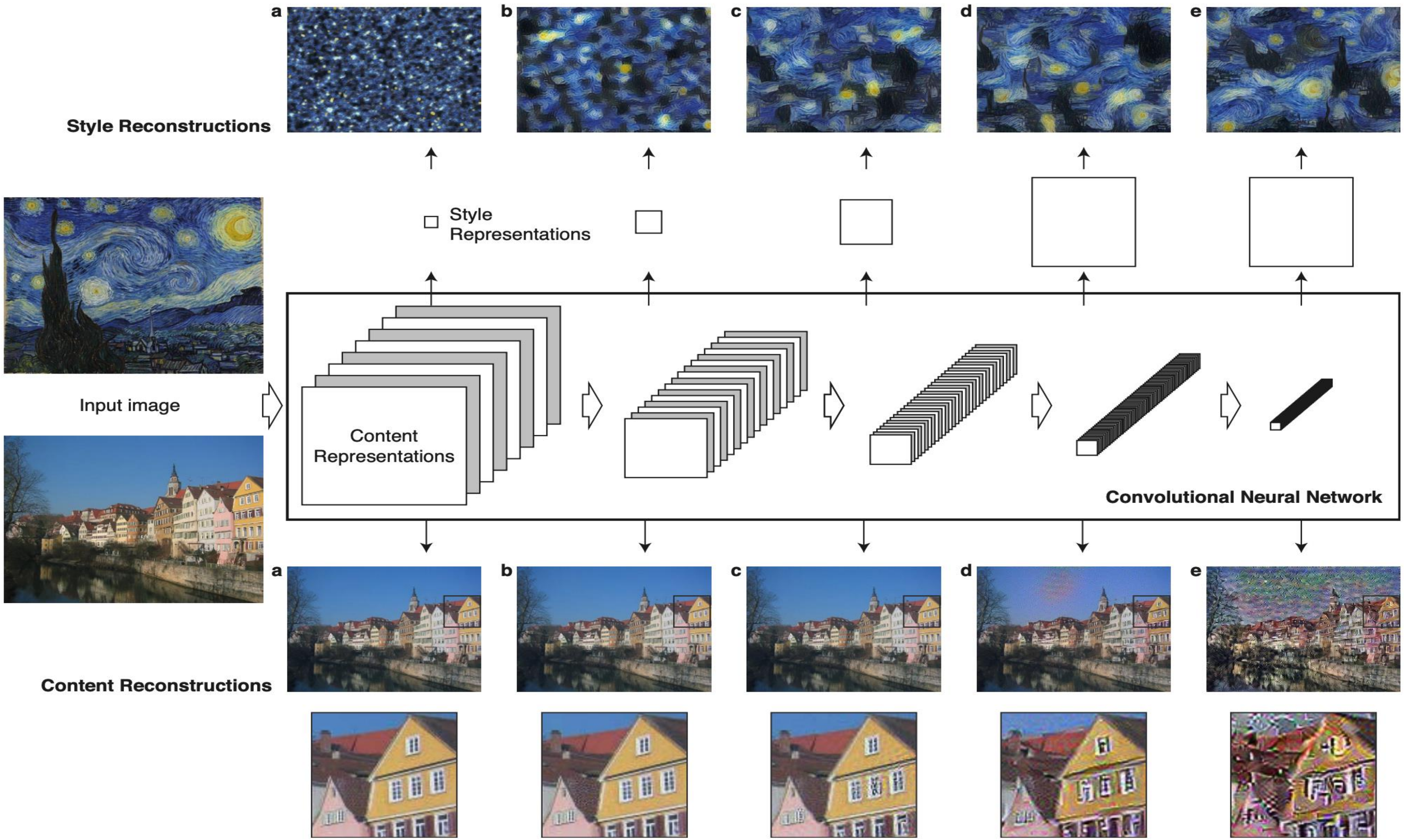
Generative Models: SRGAN & EnhanceNet & ESRGAN

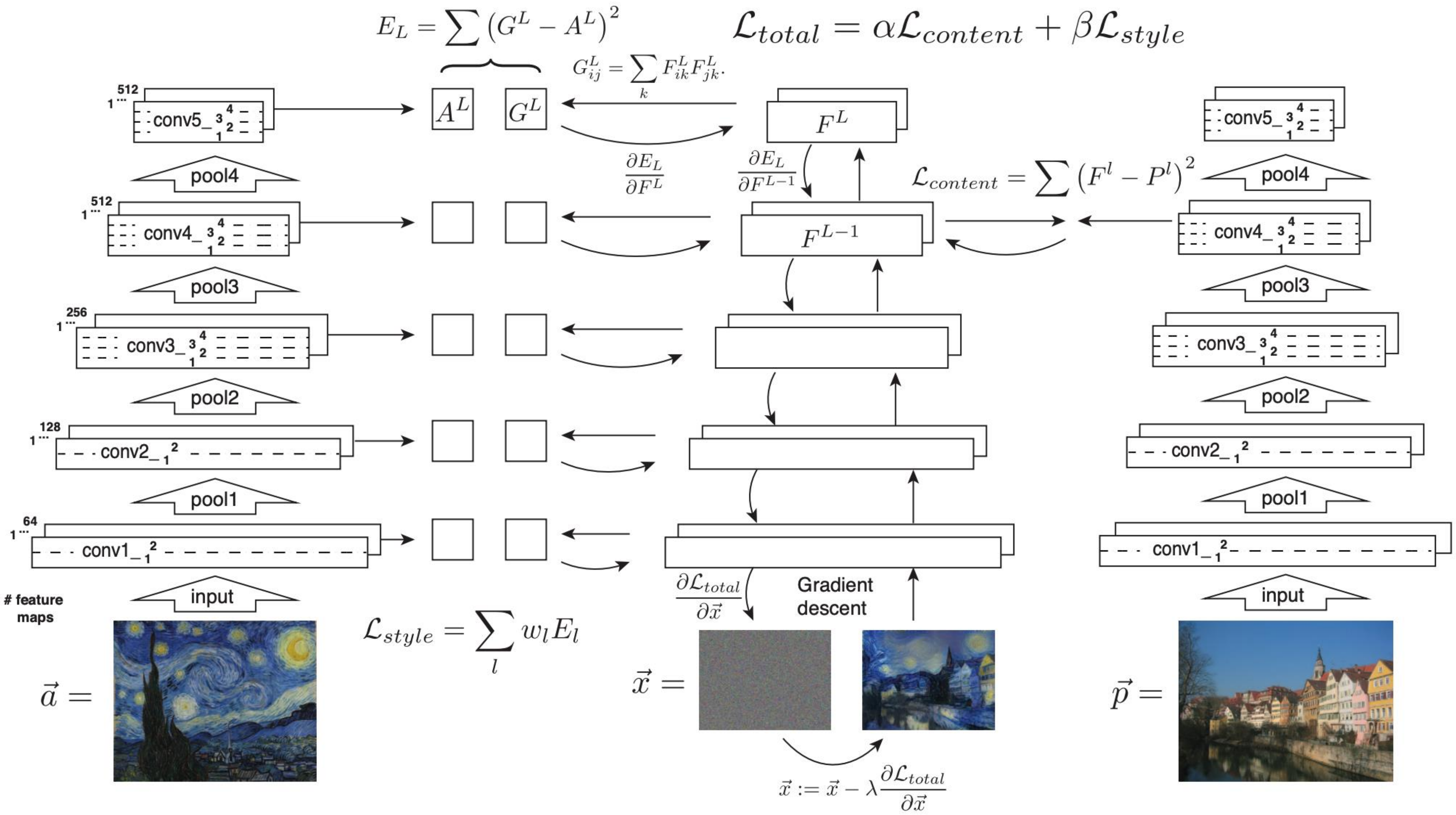
[1] Leon A. Gatys, Alexander S. Ecker, Matthias Bethge, Image Style Transfer Using Convolutional Neural Networks (CVPR 2016)

[2] Justin Johnson, Alexandre Alahi, Li Fei-Fei, Perceptual Losses for Real-Time Style Transfer and Super-Resolution (ECCV 2016)

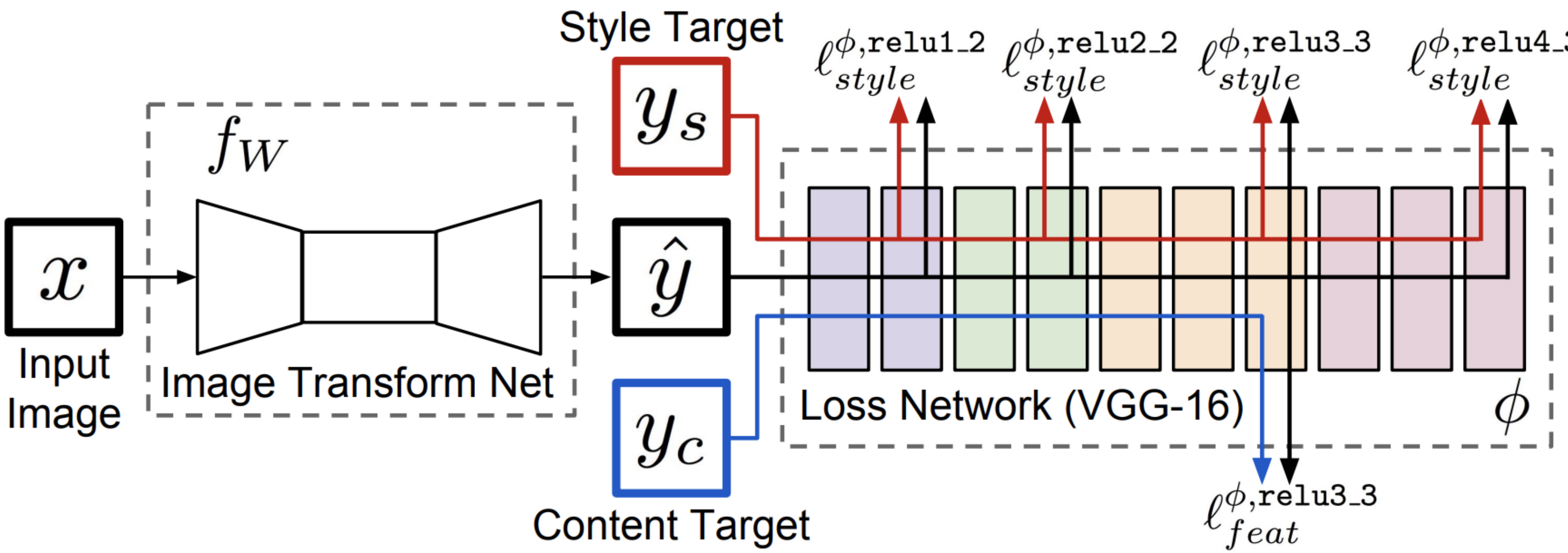
Perceptual Loss

- Feature Reconstruction Loss. Rather than encouraging the pixels of the output image ˆy = fW (x) to exactly match the pixels of the target image y, we instead encourage them to have similar feature representations as computed by the loss network φ.
- The networks discussed so far give equal importance to all spatial locations . In all our experiments φ is the 16-layer VGG network [46] pretrained on the ImageNet dataset.


Style Reconstructions
a
b
c
d
e
Style
Representations
Input image
Content
Representations
Convolutional Neural Network
Content Reconstructions
a
b
c
d
e


$E_L = \sum (G^L - A^L)^2$
$\mathcal{L}_{total} = \alpha\mathcal{L}_{content} + \beta\mathcal{L}_{style}$
$G_{ij}^L = \sum_k F_{ik}^L F_{jk}^L.$
A^L
G^L
F^L
$\frac{\partial E_L}{\partial F^L}$
$\frac{\partial E_L}{\partial F^{L-1}}$
F^{L-1}
$\mathcal{L}_{content} = \sum (F^l - P^l)^2$
512
conv5_
pool4
conv4_
pool3
256
conv3_
pool2
128
conv2_
pool1
64
conv1_
feature maps
input
$\frac{\partial \mathcal{L}_{total}}{\partial \vec{x}}$
Gradient descent
$\mathcal{L}_{style} = \sum_l w_l E_l$
$\vec{a} =$
$\vec{x} =$
$\vec{p} =$
$\vec{x} := \vec{x} - \lambda \frac{\partial \mathcal{L}_{total}}{\partial \vec{x}}$

Perceptual Loss



Feature Reconstruction Loss

$$\ell_{feat}^{\phi,j}(\hat{y}, y) = \frac{1}{C_j H_j W_j} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2$$

Style Reconstruction Loss.

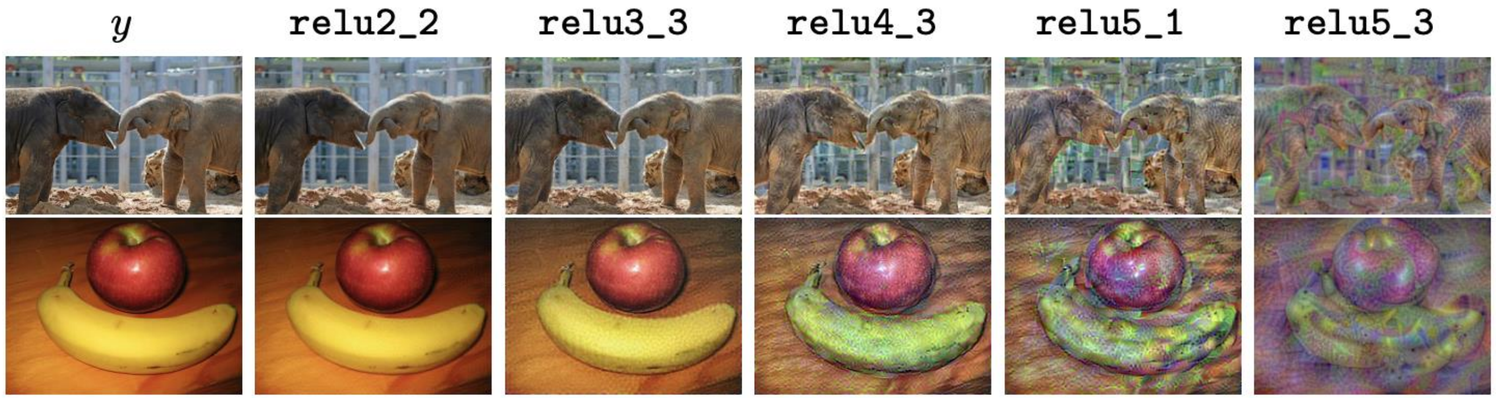
$$\ell_{style}^{\phi,j}(\hat{y}, y) = \|G_j^\phi(\hat{y}) - G_j^\phi(y)\|_F^2.$$

$$G_j^\phi(x)_{c,c'} = \frac{1}{C_j H_j W_j} \sum_{h=1}^{H_j} \sum_{w=1}^{W_j} \phi_j(x)_{h,w,c} \phi_j(x)_{h,w,c'}.$$

Perceptual Loss

Feature Reconstruction Loss

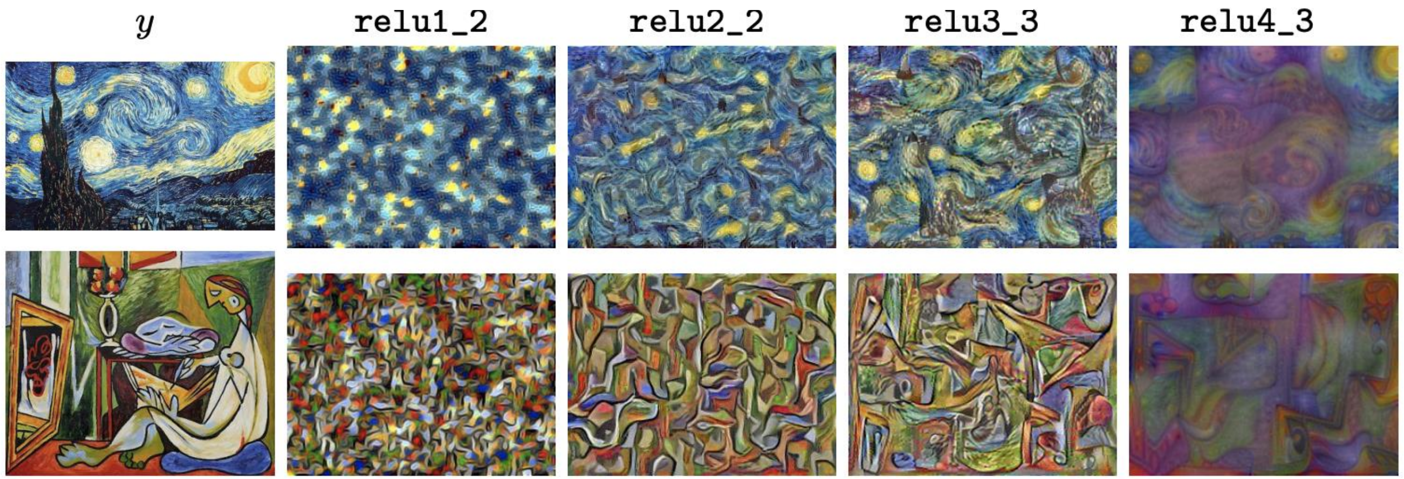
$$\ell_{feat}^{\phi,j}(\hat{y},y) = \frac{1}{C_jH_jW_j}\|\phi_j(\hat{y}) - \phi_j(y)\|_2^2$$



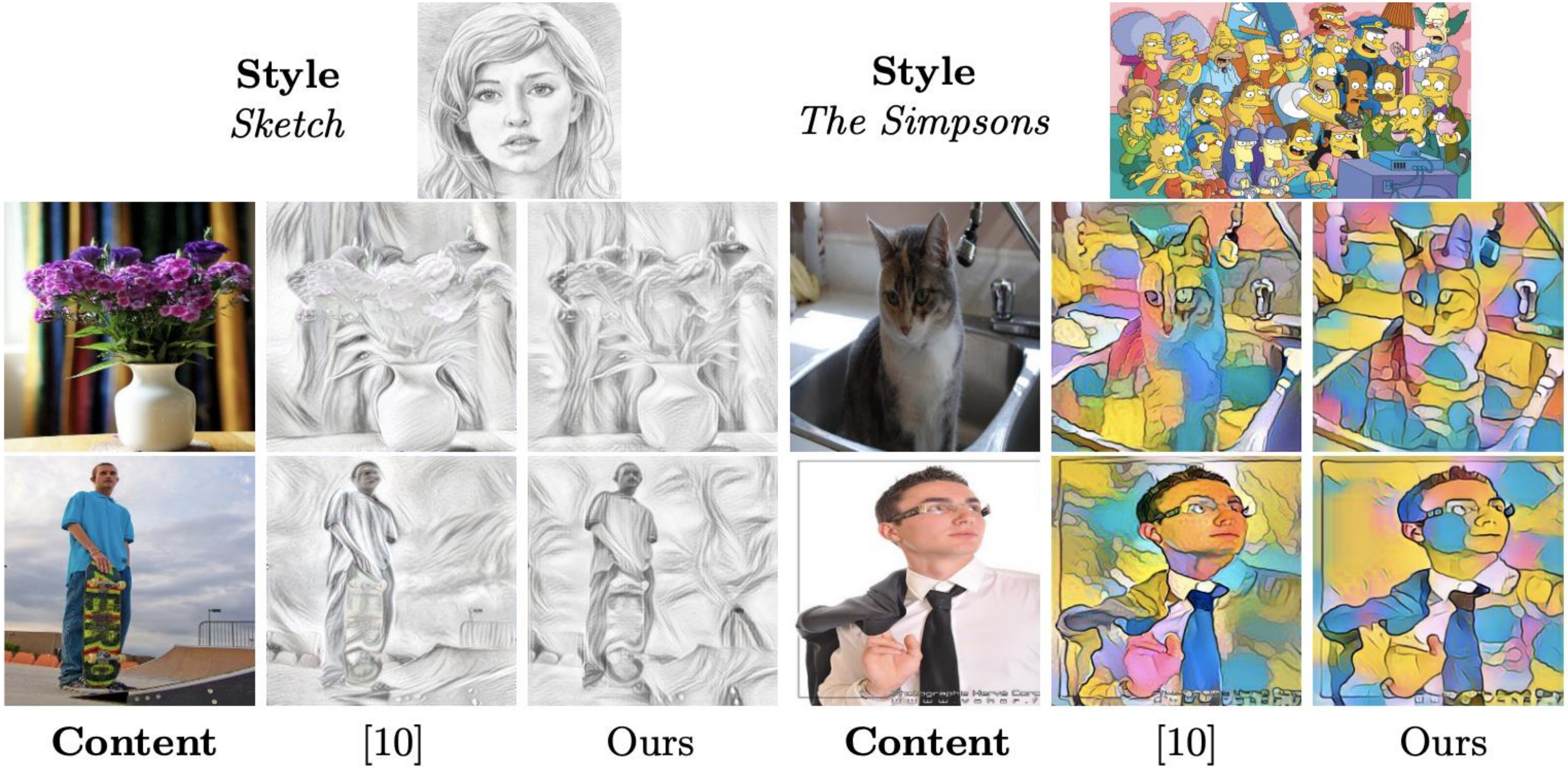
Style Reconstruction Loss.

$$\ell_{style}^{\phi,j}(\hat{y},y) = \|G_j^{\phi}(\hat{y}) - G_j^{\phi}(y)\|_F^2.$$

$$G_j^{\phi}(x) = \psi\psi^T/C_jH_jW_j.$$



Perceptual Loss



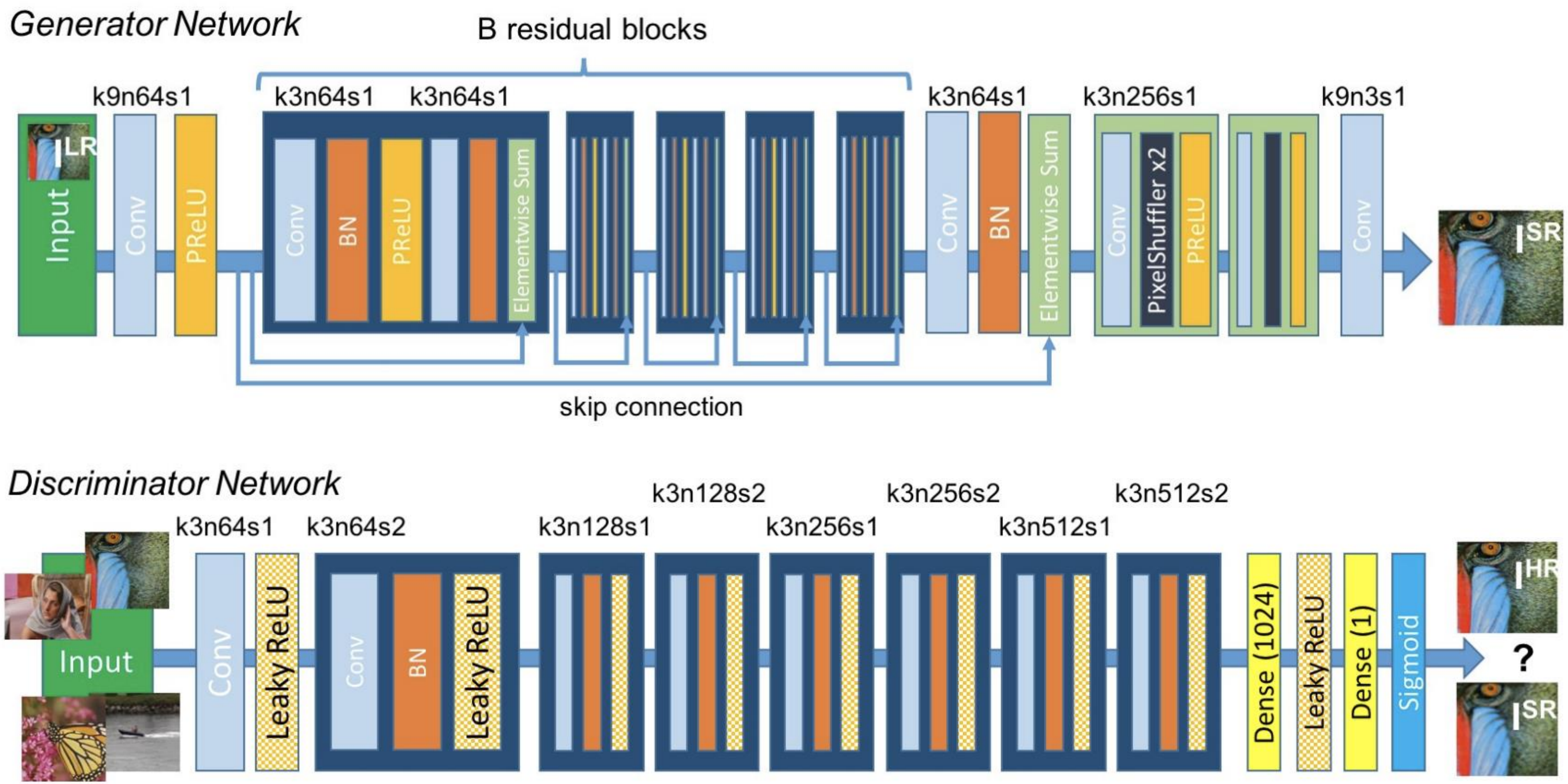
Perceptual Loss

Ground Truth	Bicubic	Ours (ℓ_{pixel})	SRCNN [11]	Ours (ℓ_{feat})
This Image	21.69 / 0.5840	21.66 / 0.5881	22.53 / 0.6524	21.04 / 0.6116
Set14 mean	25.99 / 0.7301	25.75 / 0.6994	27.49 / 0.7503	24.99 / 0.6731
BSD100 mean	25.96 / 0.682	25.91 / 0.6680	26.90 / 0.7101	24.95 / 63.17

SRGAN

- It is the first framework capable of inferring photo-realistic natural images for $4\times$ upscaling factors.
- We propose a perceptual loss function which consists of an adversarial loss and a content loss. The adversarial loss pushes our solution to the natural image manifold using a discriminator network that is trained to differentiate between the super-resolved images and original photo-realistic images

SRGAN



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SRGAN

bicubic (21.59dB/0.6423) | SRResNet (23.53dB/0.7832) | SRGAN (21.15dB/0.6868) | original

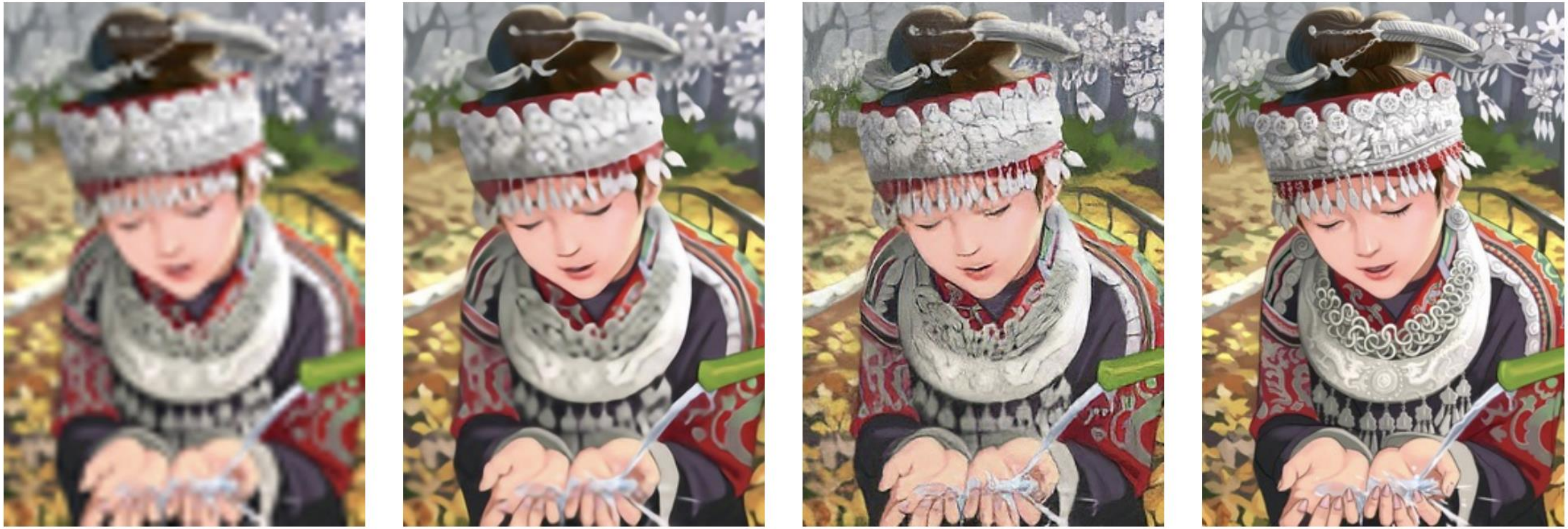


Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]

SRGAN

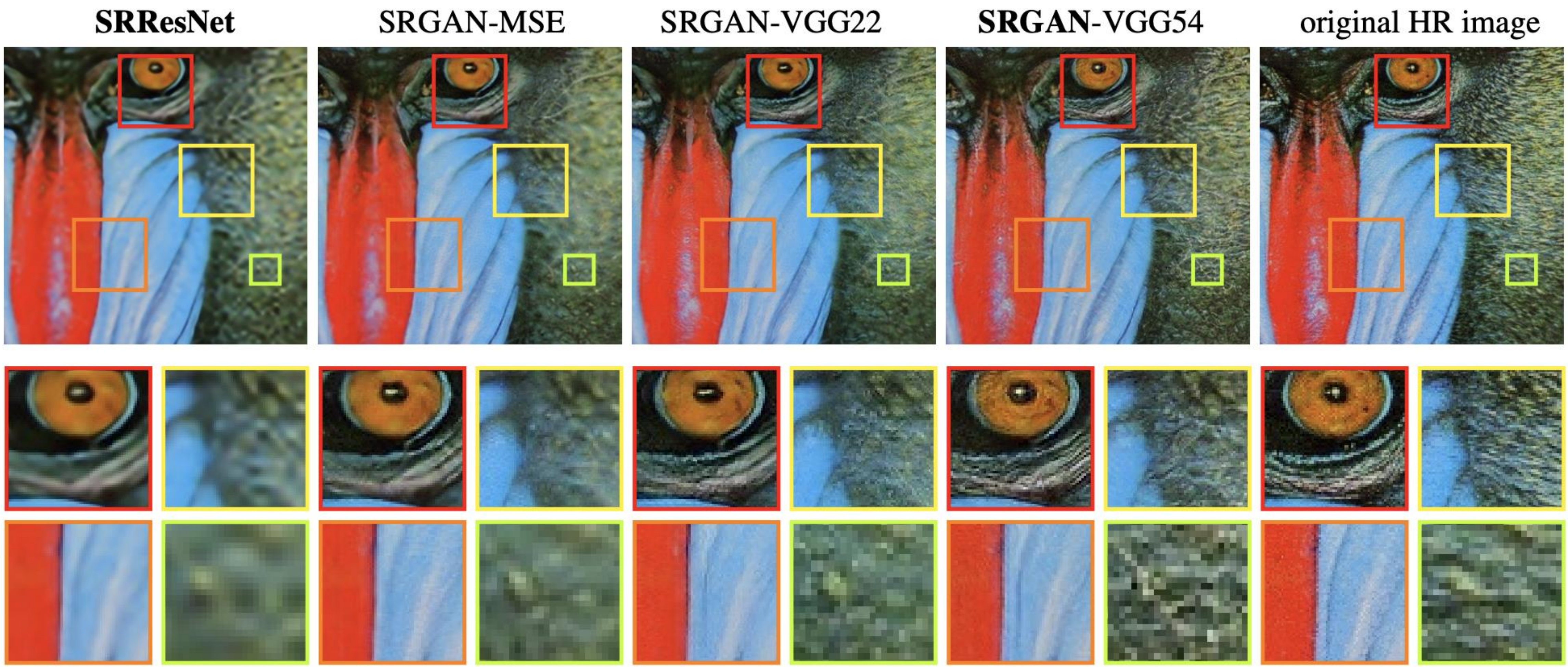


Figure 6: **SRResNet** (left: a,b), SRGAN-MSE (middle left: c,d), SRGAN-VGG2.2 (middle: e,f) and **SRGAN**-VGG54 (middle right: g,h) reconstruction results and corresponding reference HR image (right: i,j). [4× upscaling]

Summary

Image Super-resolution

Supervised Image Super-resolution

- **Model Frameworks**
  - Pre-upsampling SR
  - Post-upsampling SR
  - Progressive Upsampling SR
  - Iterative Up-and-down Sampling SR
- **Upsampling Methods**
  - Interpolation-based Methods:
    - Nearest Neighbor
    - Bilinear
    - Bicubic
    - Others
  - Learning-based Methods:
    - Transposed Convolution
    - Sub-pixel Layer
    - Meta Upscale Module
- **Network Design**
  - Residual Learning
  - Recursive Learning
  - Multi-path Learning
  - Dense Connections
  - Attention Mechanism
  - Advanced Convolution
  - Region-recursive Learning
  - Pyramid Pooling
  - Wavelet Transformation
  - xUnit
  - Desubpixel
- **Learning Strategies**
  - Loss Functions:
    - Pixel Loss
    - Content Loss
    - Texture Loss
    - Adversarial Loss
    - Cycle Consistency Loss
    - Total Variation Loss
    - Prior-based Loss
  - Batch Normalization
  - Curriculum Learning
  - Multi-supervision
- **Other Improvements**
  - Context-wise Network Fusion
  - Data Augmentation
  - Multi-task Learning
  - Network Interpolation
  - Self-ensemble

Performance Evaluation

- Benchmark Datasets
- Performance Metric:
  - Objective Methods: PSNR, SSIM, etc.
  - Subjective Methods: MOS
  - Task-based Evaluation
  - Learning-based Perceptual Quality
  - Other Methods
- Operating Channels

Unsupervised Image Super-resolution

- Zero-shot Super-resolution
- Weakly-supervised Super-resolution:
  - Learned Degradation
  - Cycle-in-cycle Super-resolution
- Deep Image Prior

Domain-specific Applications

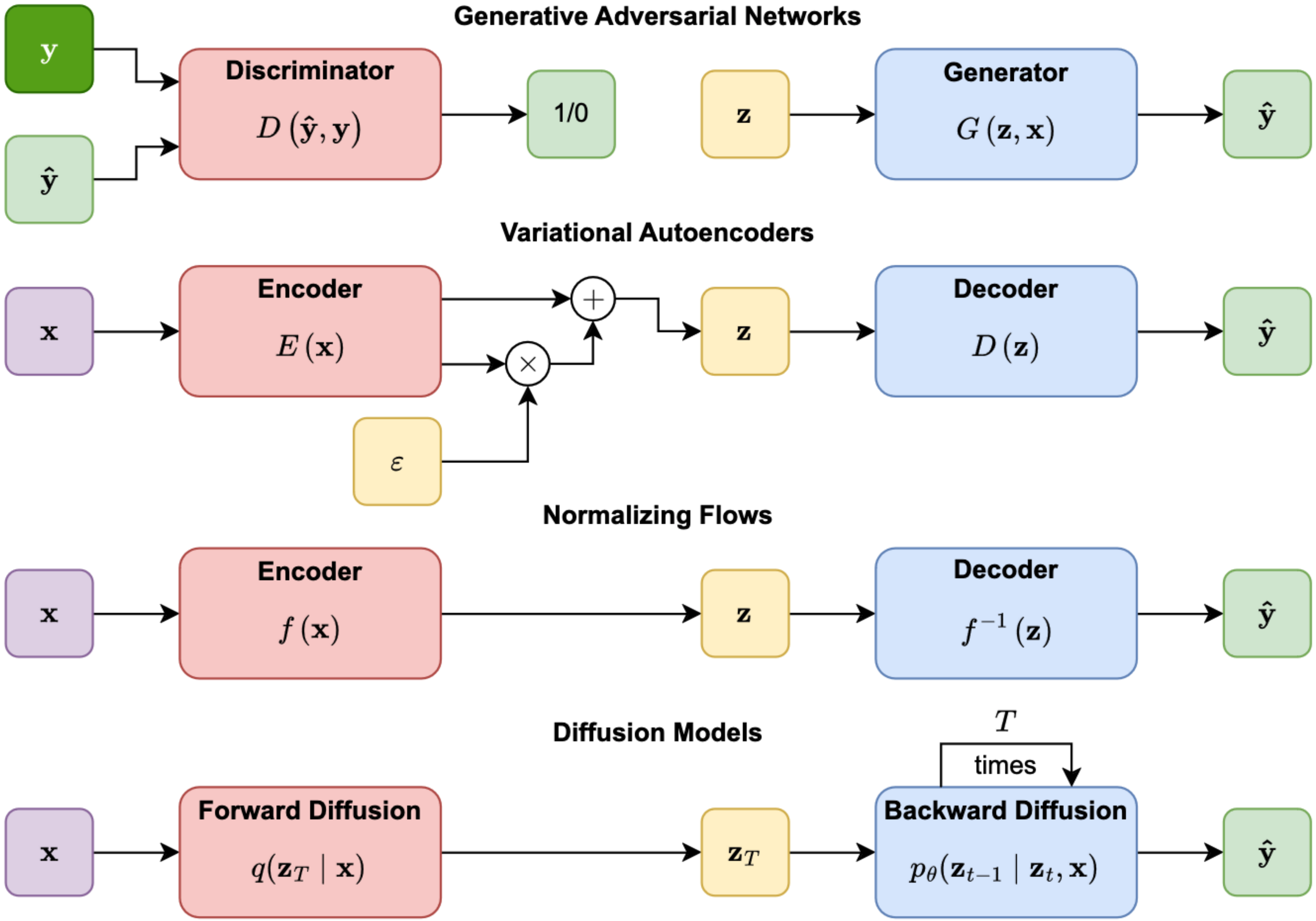
- Depth Map Super-resolution
- Face Image Super-resolution
- Hyperspectral Image Super-resolution
- Real-world Image Super-resolution
- Video Super-resolution
- Other Applications

Diffusion Model based Super Resolution

[1] SR3: Image Super-Resolution via Iterative Refinement, TPAMI 2023

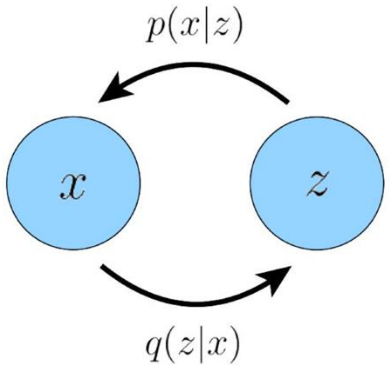
[2] Implicit Diffusion Models for Continuous Super-Resolution, CVPR 2023

Conceptual overview of generative models (GANs, VAEs, NFs, and DMs).



VAE-Variational Autoencoders

$$
\begin{aligned}
\log p(\boldsymbol{x}) &= \log p(\boldsymbol{x}) \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) dz && \text{(Multiply by } 1 = \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) d\boldsymbol{z}\text{)} \\
&= \int q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) (\log p(\boldsymbol{x})) dz && \text{(Bring evidence into integral)} \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} [\log p(\boldsymbol{x})] && \text{(Definition of Expectation)} \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{p(\boldsymbol{z}|\boldsymbol{x})}\right] && \text{(Apply Equation 2)} \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z}) q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{p(\boldsymbol{z}|\boldsymbol{x}) q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] && \text{(Multiply by } 1 = \frac{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\text{)} \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}{p(\boldsymbol{z}|\boldsymbol{x})}\right] && \text{(Split the Expectation)} \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}|\boldsymbol{x})) && \text{(Definition of KL Divergence)} \\
&\geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\right] && \text{(KL Divergence always} \geq 0\text{)}
\end{aligned}
$$



Hierarchical VAE

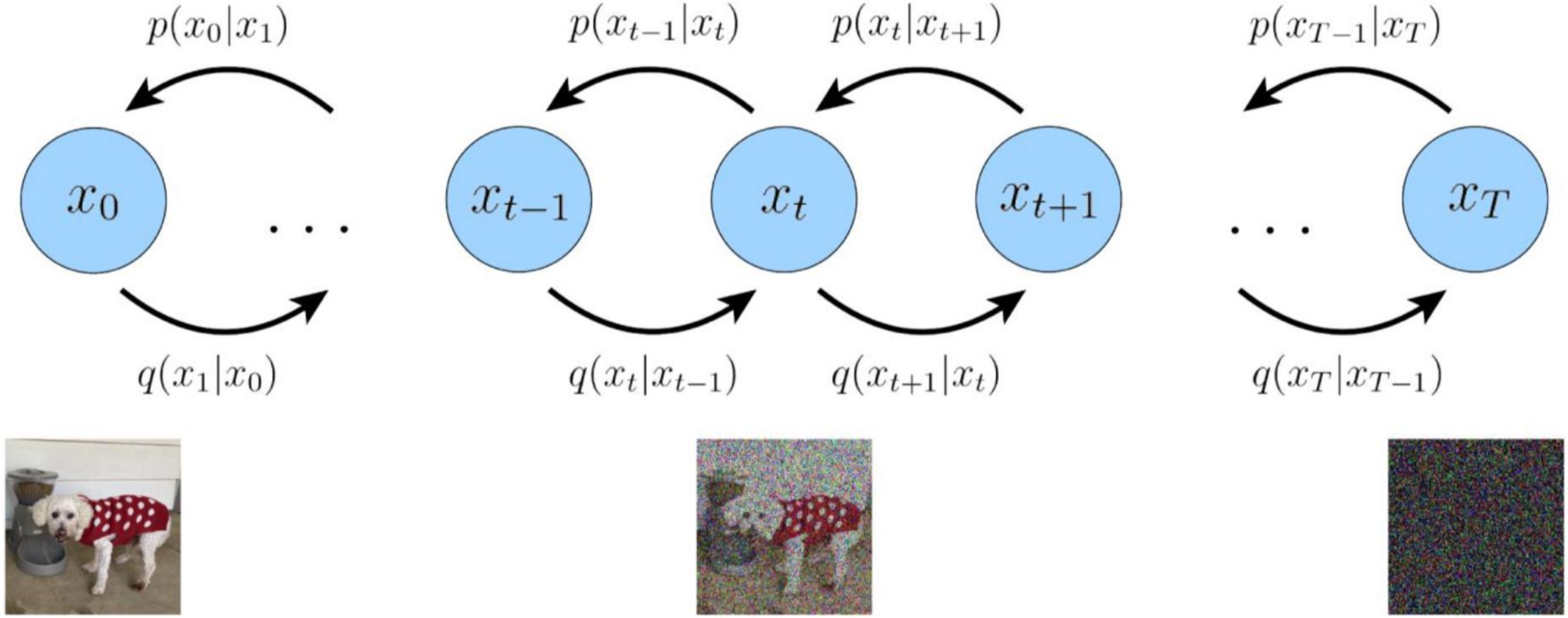
This MHVAE satisfies the following three constraints (note that these three constraints also form the foundation of the entire diffusion model inference):

1. The dimensionality of the latent vector Z is consistent with the dimensionality of the input X.
2. The latent vector at each time step is encoded as a Gaussian distribution that depends only on the latent vector of the previous time step.
3. The parameters of the Gaussian distribution for the latent vector at each time step vary with the time step, and the Gaussian distribution at the final time step satisfies the constraint of being a standard Gaussian distribution.

Hierarchical VAE

1. Dimension consistency; 2. Gaussian distribution encoding

3. With multiple time step, end with a standard Gaussian distribution



Hierarchical VAE

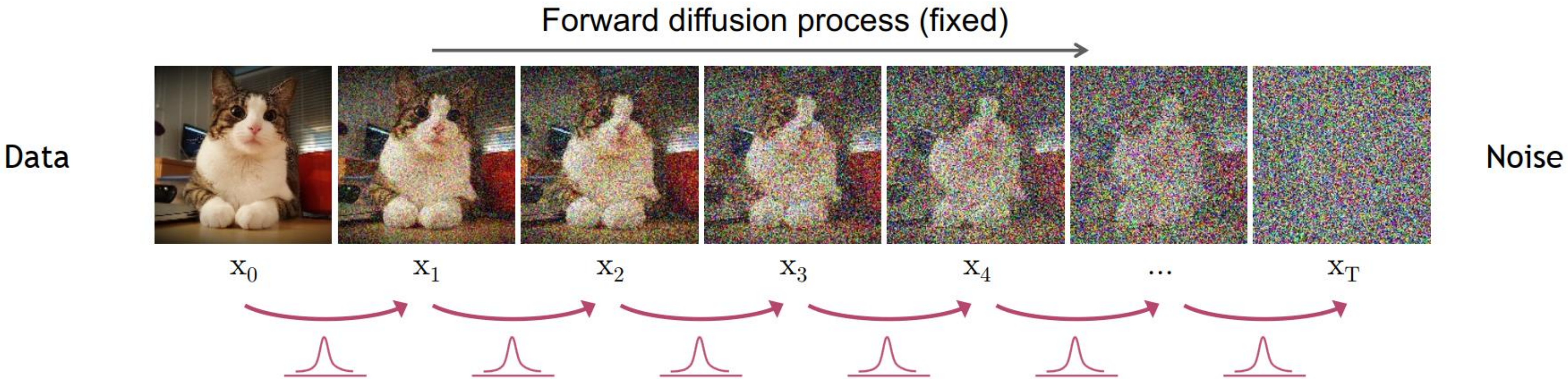
1. Dimension consistency； 2. Gaussian distribution encoding

3. With multiple time step， end with a standard Gaussian distribution

$$
\begin{aligned}
\log p(\boldsymbol{x}) &= \log \int p(\boldsymbol{x}, \boldsymbol{z}_{1:T}) d\boldsymbol{z}_{1:T} \\
&= \log \int \frac{p(\boldsymbol{x}, \boldsymbol{z}_{1:T}) q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})} d\boldsymbol{z}_{1:T} && \left(\text{Multiply by } 1 = \frac{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})}\right) \\
&= \log \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})} \left[\frac{p(\boldsymbol{x}, \boldsymbol{z}_{1:T})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})} \right] && (\text{Definition of Expectation}) \\
&\geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{x}, \boldsymbol{z}_{1:T})}{q_{\boldsymbol{\phi}}(\boldsymbol{z}_{1:T}|\boldsymbol{x})} \right] && (\text{Apply Jensen's Inequality})
\end{aligned}
$$

Forward Diffusion Process

The formal definition of the forward process in T steps

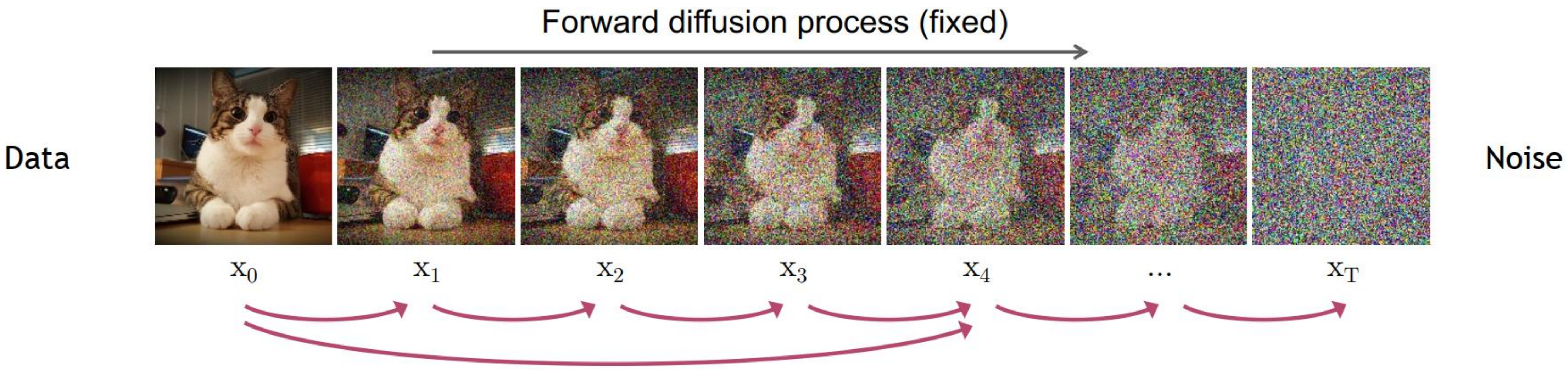


$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad \longrightarrow \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

The steps sizes are controlled by a variance schedule $\{\beta_t \in (0,1)\}_{t=1}^{T}$

Diffusion Kernel

The formal definition of the forward process in T steps



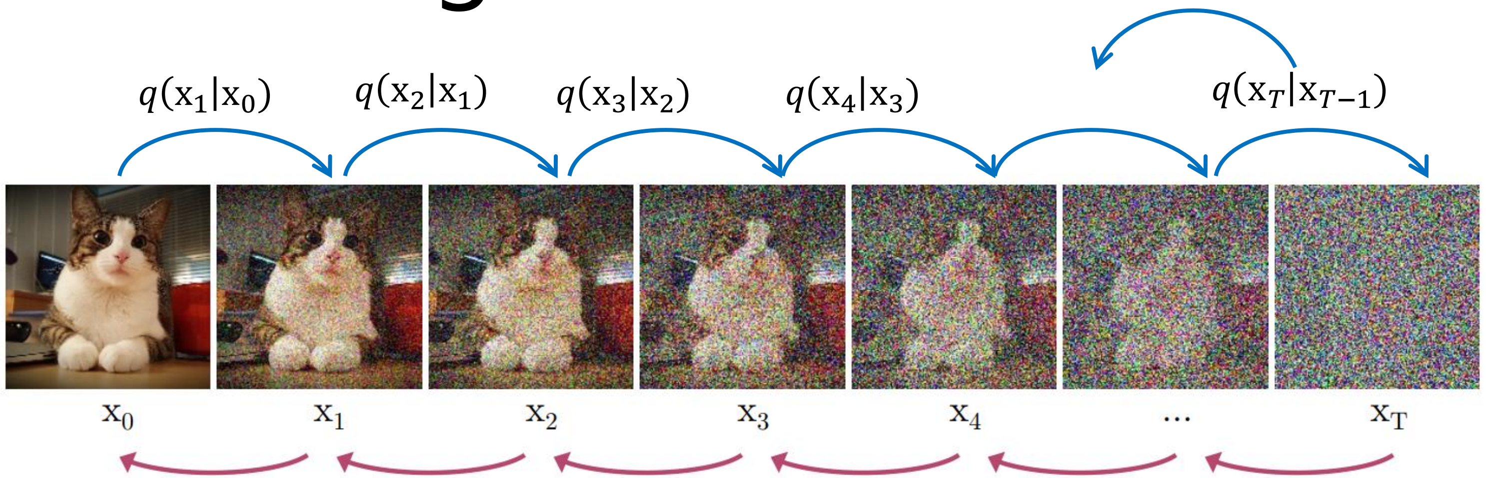
Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha} = \prod_{i=1}^{t} \alpha_i$:

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} && ;\text{where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\
&= \sqrt{\alpha_t \alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t \alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2} && ;\text{where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians } (*). \\
&= \dots \\
&= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}
\end{aligned}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$ Diffusion Kernel

(*)
$\epsilon_1 \in N(0, \sigma_1^2 \boldsymbol{I})$
$\epsilon_2 \in N(0, \sigma_2^2 \boldsymbol{I})$
$\epsilon_1 + \epsilon_2 \in N(0, (\sigma_1^2 + \sigma_2^2)\boldsymbol{I})$

Reverse Denoising Process



Train a network to approximate $q(x_{t-1}|x_t)$

$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$
$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\mu_\theta(\mathbf{x}_t, t)}_{\text{Trainable network (U-net, Denoising Autoencoder)}}, \sigma_t^2\mathbf{I})$$

$$\Rightarrow \quad p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

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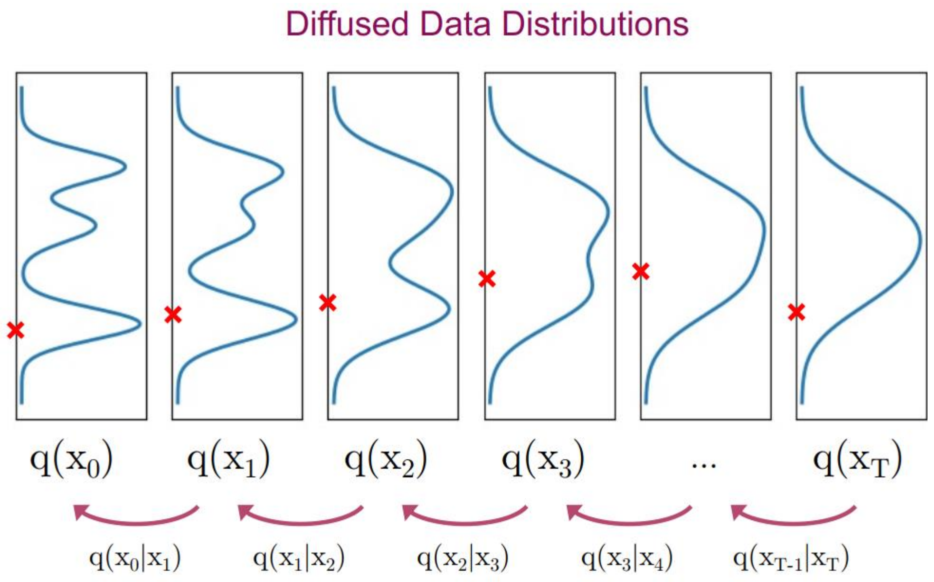
Generative Learning by Denoising

We know prior $q(\mathrm{x}_t|\mathrm{x}_{t-1})$, Can we get posterior $q(\mathrm{x}_{t-1}|\mathrm{x}_t)$?



In general, $q(\mathrm{x}_{t-1}|\mathrm{x}_t) \propto q(\mathrm{x}_{t-1})q(\mathrm{x}_t|\mathrm{x}_{t-1})$ is intractable

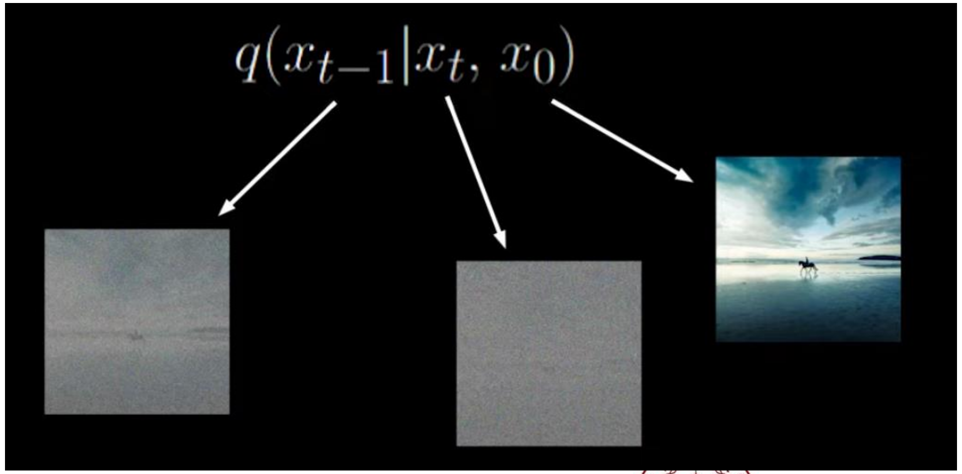
Note that when β_t is small enough, $q(x_{t-1}|x_t)$ will be Gaussian

But we cannot easily estimate $q(x_{t-1}|x_t)$ it **needs entire dataset.**

Generation:

Sample $\mathrm{x}_T \sim N(\mathrm{x}_T; 0, \mathrm{I})$

Iteratively sample $\mathrm{x}_{t-1} \sim q(\mathrm{x}_{t-1}|\mathrm{x}_t)$



$q(x_{t-1}|x_t)$ is intractable; $q(x_{t-1}|x_t, x_0)$ is tractable

Conditional forward and reverse process

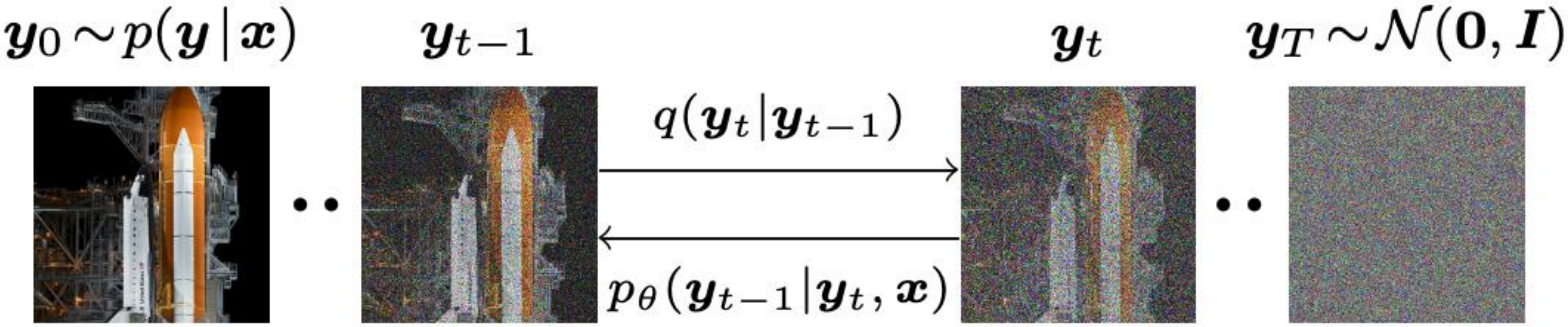


Figure 2: The forward diffusion process q (left to right) gradually adds Gaussian noise to the target image. The reverse inference process p (right to left) iteratively denoises the target image conditioned on a source image $\boldsymbol{x}$. Source image $\boldsymbol{x}$ is not shown here.

SR performance of SR3

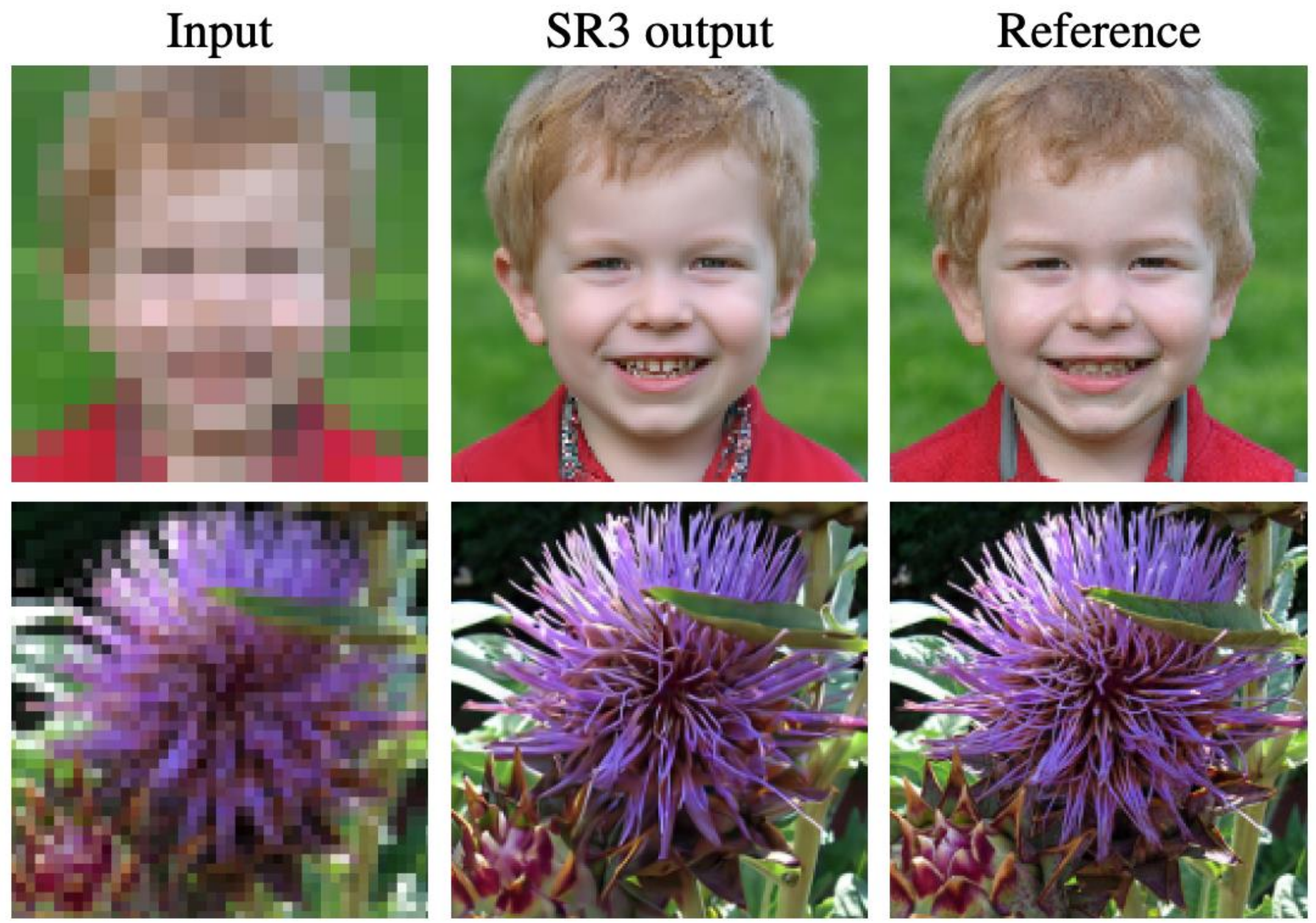
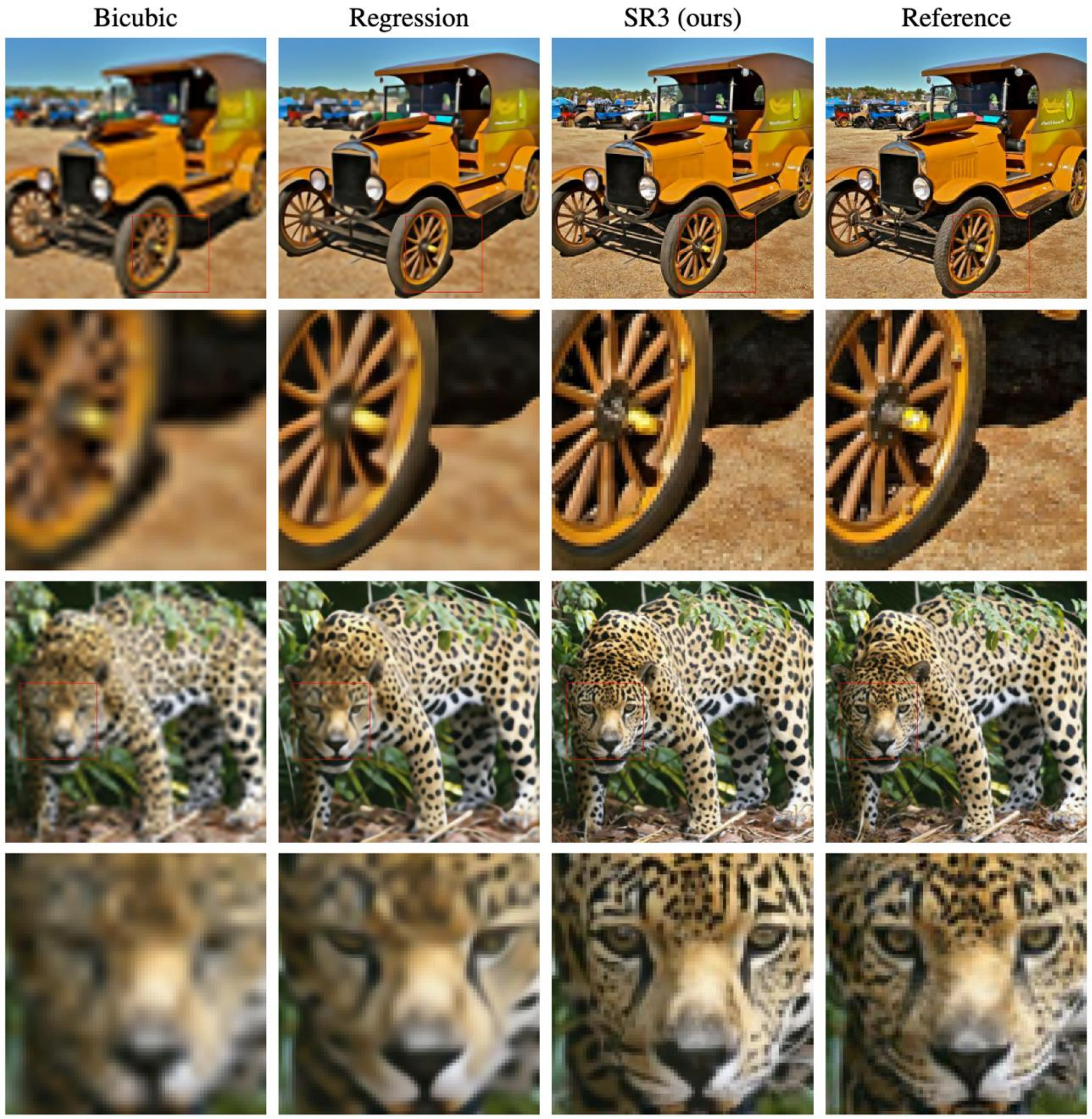
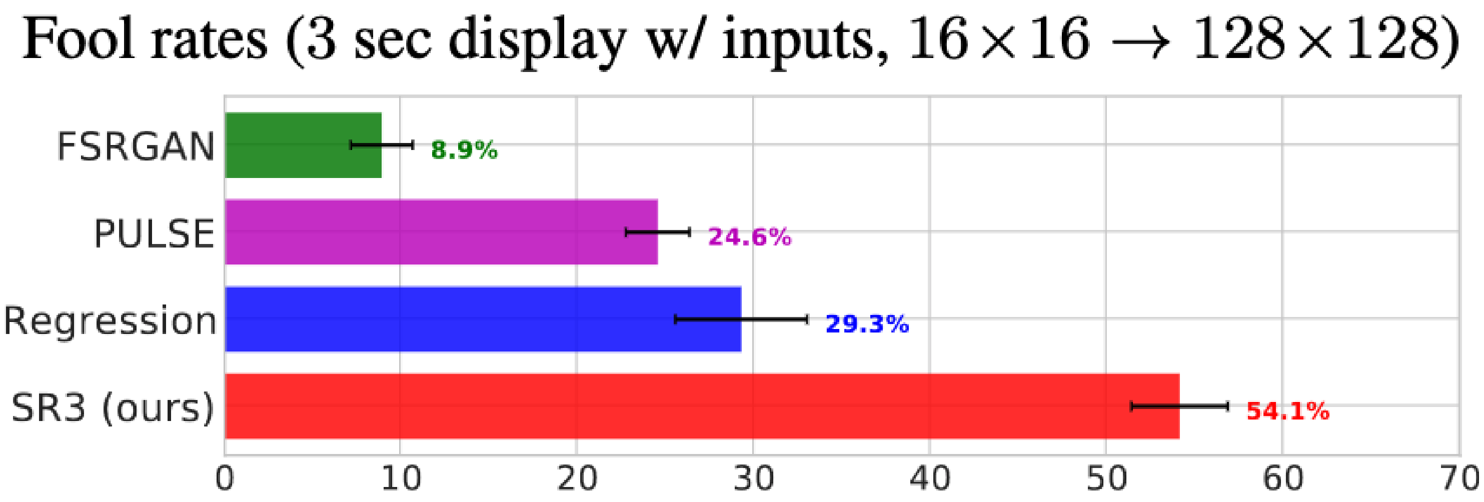
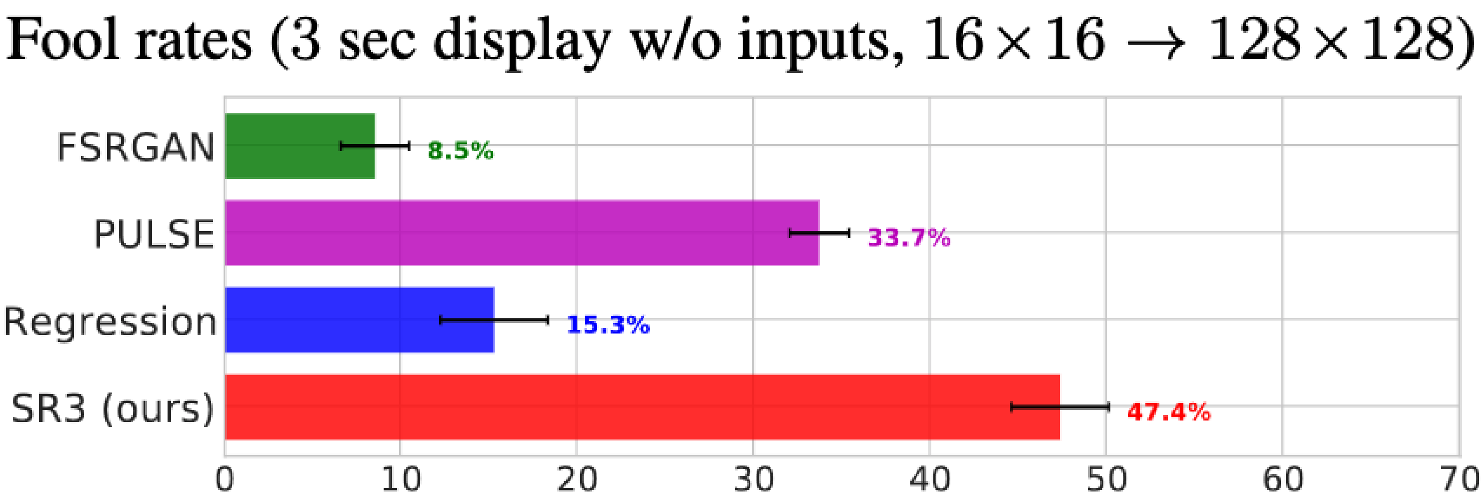


Figure 1: Two representative SR3 outputs: (top) $8\times$ face super-resolution at $16\times16\to128\times128$ pixels (bottom) $4\times$ natural image super-resolution at $64\times64\to256\times256$ pixels.



SR performance of SR3





IDM: Implicit Diffusion Models for Continuous Super-Resolution

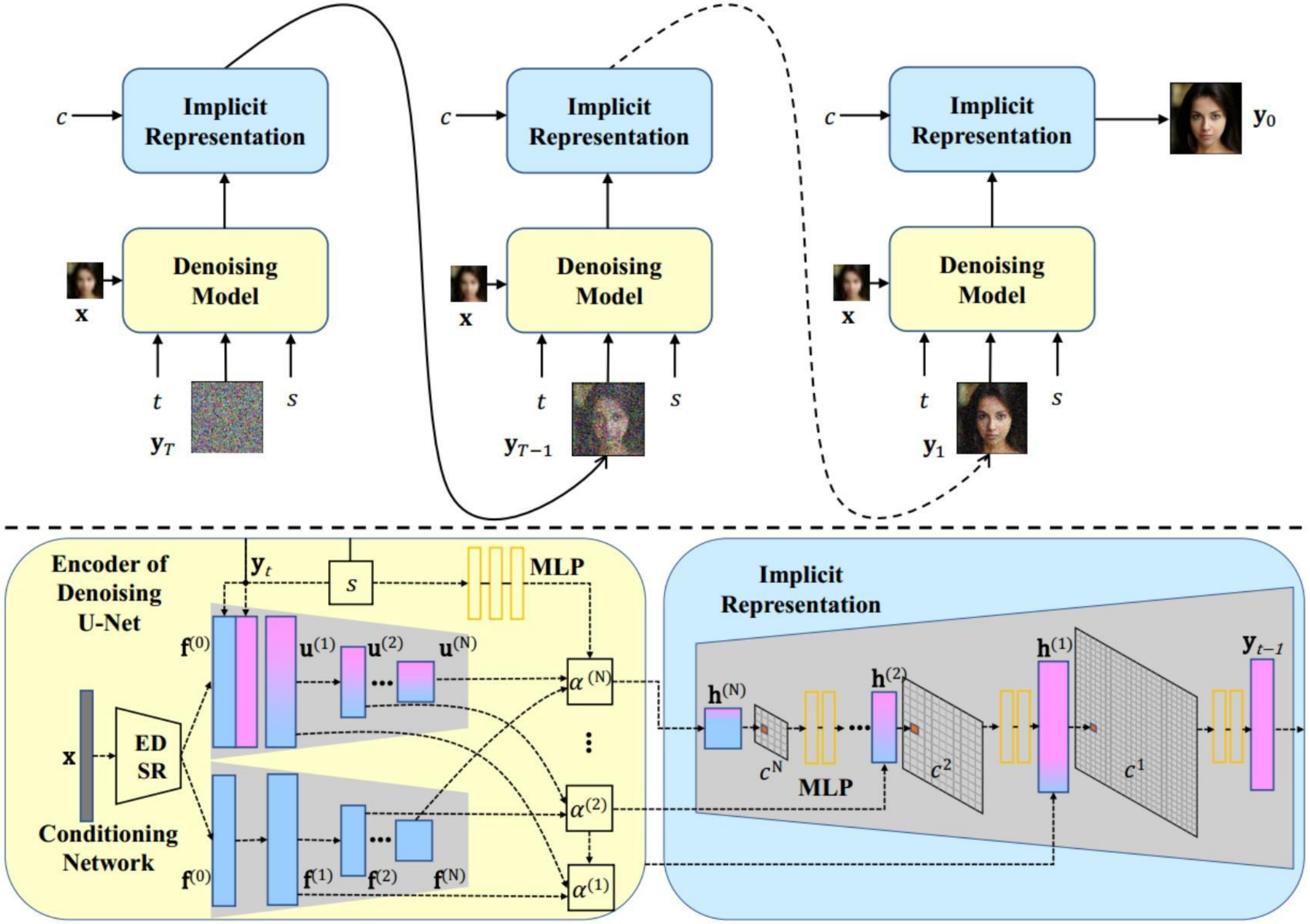


Figure 3. Overview of the IDM framework. **Upper Part:** Overall process of the inference. **Lower Part:** Detailed illustration of a denoising step, where the U-Net decoder is omitted for conciseness.

Performance of IDM

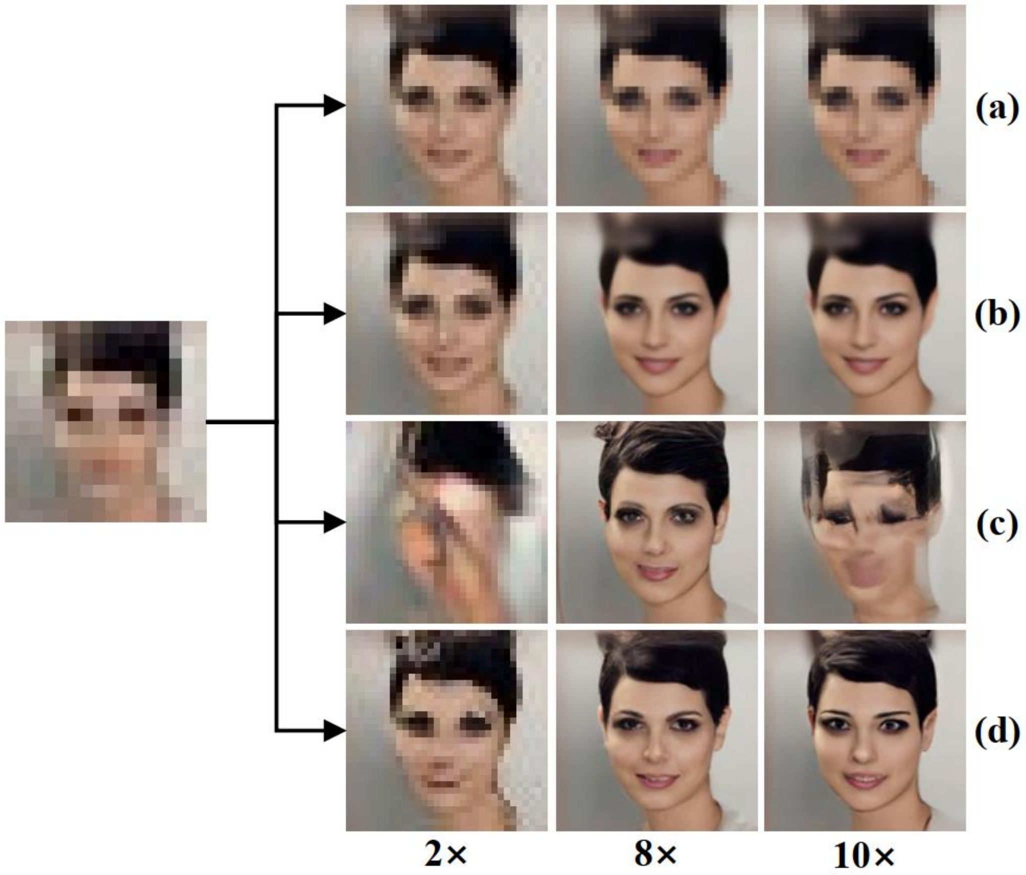
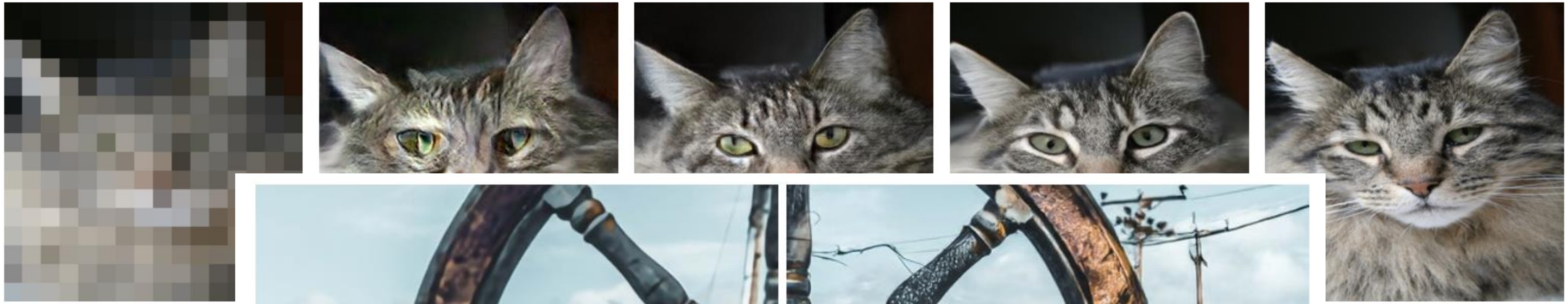


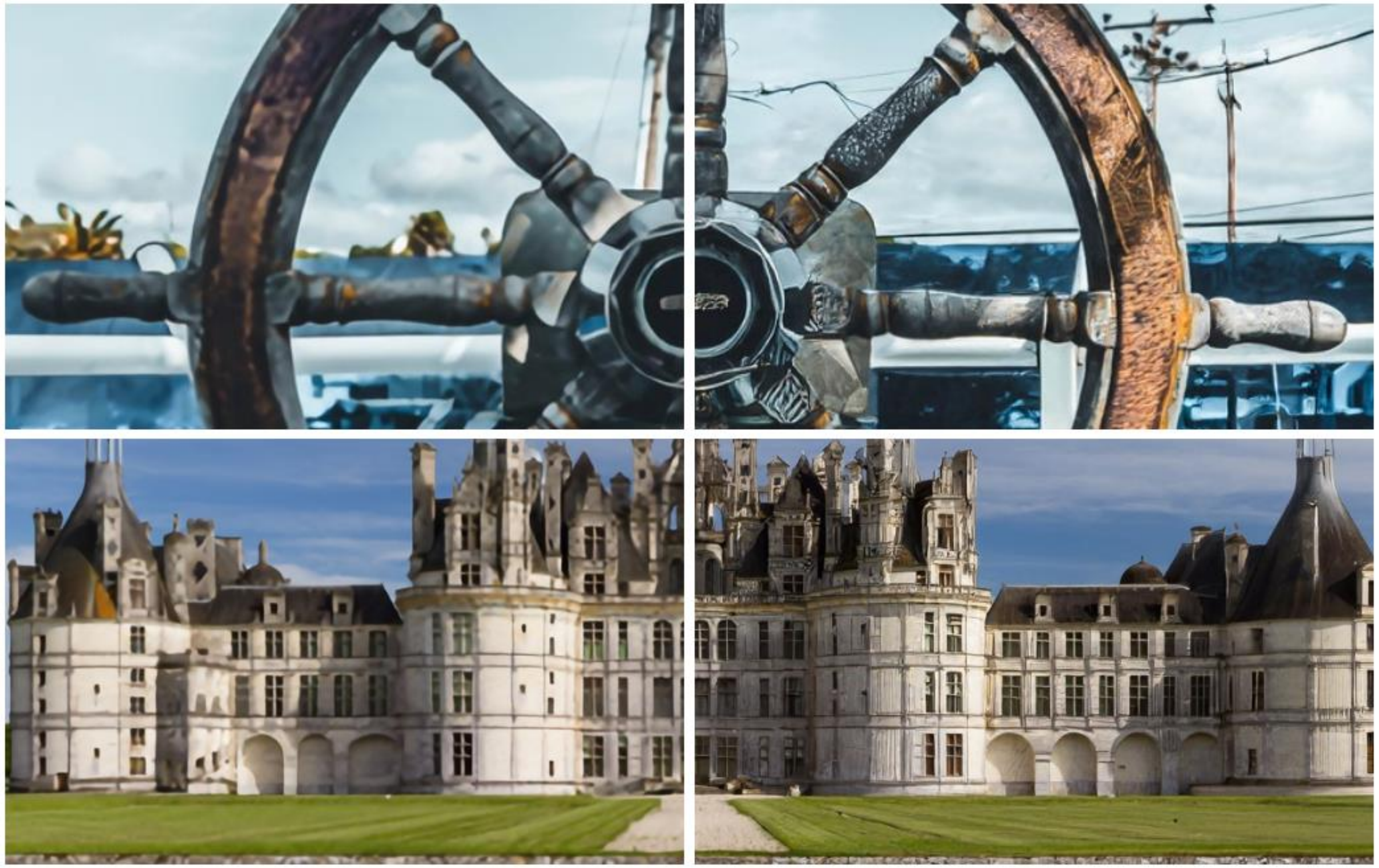
Figure 1. Visual comparison, where training is on 8× SR and testing on 2×, 8×, and 10×. (a) EDSR [23] and (b) LIIF [6] are regression-based models; (c) SR3 [35] and (d) IDM (ours) are generative models. Among them, LIIF and IDM employ the implicit neural representation.

Performance of IDM



(a) Low-Resolu ...) Ground-Truth

Figure 2. Examples ... re GAN, and loses the
inherent information. ... ill generates unnatural
textures and GAN-spe ... anism, IDM generates
the output with high-f



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Take home message

- Learning high-frequency image content is difficult for supervised neural network paradigm. There are two ways to improve:
- Unsupervised methods to overfit the high-frequency image content in anisotropic image.
- Generative methods learn the latent space consistency between low-resolution and high-resolution image, and avoid from generating perceptual incorrect high-frequency texture.